

Pin Connections & Wiring

Arduino Uno ↔ Ultrasonic Sensor HC-SR04

HC-SR04 Pin Arduino Uno Pin

VCC	5V
GND	GND
TRIG	A0
ECHO	A1

Where Analog pins used as digital I/O

◆ Arduino Uno ↔ L298N Motor Driver

L298N Pin	Arduino Uno Pin	Function
ENA	D10	Motor A speed (PWM)
IN1	D9	Motor A direction
IN2	D8	Motor A direction
IN3	D7	Motor B direction
IN4	D6	Motor B direction
ENB	D5	Motor B speed (PWM)
GND	GND	Common ground

⚠ Remove the 5V_EN jumper on the L298N

Motors & L298N Connections

Motor L298N Terminal

Left Motor	OUT1 & OUT2
Right Motor	OUT3 & OUT4

Power Connections

2×18650 Li-ion Battery Pack (7.4V)

Battery Connection

Battery +	L298N +12V (VMOTOR)
Battery –	L298N GND

Buck Converter (Step-Down Regulator)

Input Side

Buck Converter Connection

IN+ Battery +

IN- Battery -

Output Side (5V Regulated)

Buck Converter Connection

OUT+ Arduino 5V

OUT- Arduino GND

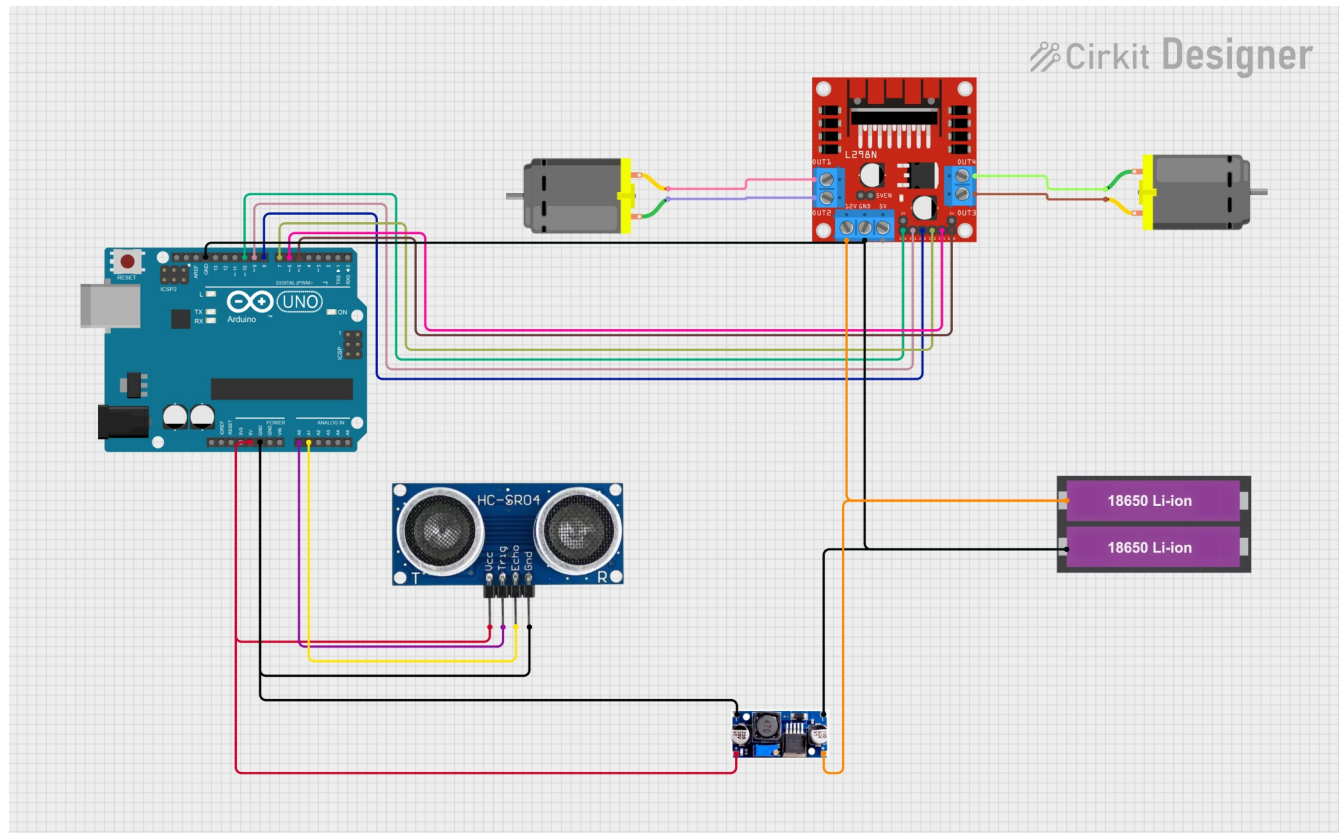
Adjust the buck converter output to exactly 5.0V before connecting to Arduino

Common Ground (Critical)

All grounds must be connected together:

- Battery GND
- Buck converter GND
- Arduino GND
- L298N GND

circuit diagram



code

```
/*  
  Obstacle Avoidance Robot  
  -----  
  Uses an ultrasonic sensor (HC-SR04) to detect obstacles  
  and controls two DC motors via L298 motor driver.  
  
  Author: Your Name  
  Date: 2026  
*/  
  
#define ENA 10      // Motor A speed control  
#define IN1 9       // Motor A direction  
#define IN2 8  
#define IN3 7       // Motor B direction  
#define IN4 6  
#define ENB 5       // Motor B speed control  
  
#define TRIG_PIN A14  
#define ECHO_PIN A15  
  
int duration;  
int distance;  
const int SAFE_DISTANCE = 30; // cm  
  
void setup() {  
  Serial.begin(9600);  
  
  // Motor pins
```

```

    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
    pinMode(IN3, OUTPUT);
    pinMode(IN4, OUTPUT);

    // Ultrasonic sensor pins
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);
}

void loop() {
    distance = measureDistance();
    Serial.println(distance);

    if (distance < SAFE_DISTANCE) {
        avoidObstacle();
    } else {
        moveForward();
    }
}

// ----- FUNCTIONS ----- //

int measureDistance() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

```

```
    duration = pulseIn(ECHO_PIN, HIGH);  
    return duration / 58.2; // Convert to cm  
}
```

```
void moveForward() {  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
    analogWrite(ENA, 80);  
  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
    analogWrite(ENB, 80);  
}
```

```
void avoidObstacle() {  
    // Move backward  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
    analogWrite(ENA, 80);  
    analogWrite(ENB, 80);  
    delay(1000);  
  
    // Stop  
    analogWrite(ENA, 0);  
    analogWrite(ENB, 0);  
    delay(500);  
  
    // Turn right
```

```
digitalWrite(IN1, HIGH);  
digitalWrite(IN2, LOW);  
digitalWrite(IN3, LOW);  
digitalWrite(IN4, HIGH);  
analogWrite(ENA, 180);  
analogWrite(ENB, 180);  
delay(800);  
}
```